Theorem. (The Capture Theorem) Let A be a nonempty subset of $\mathbb{R}$.

(a) If A is bounded above, then any open interval containing $\sup A$ contains an element of A.

(b) Similarly, if A is bounded below, then any open interval containing $\inf A$ contains an element of A.

If A is a subset of $\mathbb{R}$ which does not contain 0, we define $A^{-1} = \left\{\frac{1}{a} : a \in A\right\}$ to be the set of reciprocals of elements of A.

Theorem. (The Flipping-Huge Theorem) Let A be a set of positive real numbers. Then $\inf A = 0$ if and only if A^{-1} is not bounded above.

Proof. We need to prove both directions of

$$\inf A = 0 \iff A^{-1} \text{ is not bounded above.}$$

3. ($\Longrightarrow$) If $\inf A = 0$, suppose that A^{-1} *is* bounded above and derive a contradiction.

4. ($\Longleftarrow$) Suppose now that A^{-1} is not bounded above. Since each element of A is positive, 0 is a lower bound for A. To show that 0 is the *greatest* lower bound, suppose $\inf A > 0$ and derive a contradiction.

Exercises.

(a) Let $A = \{\frac{1}{2^n} : n \in \mathbb{N}\} = \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \dots\}$. Show that $\inf A = 0$. (*Hint*: By the Flipping Huge Theorem, you may equivalently show that $A^{-1} = \{2^n : n \in \mathbb{N}\}$ is not bounded above. Prove, by contradiction, that A^{-1} cannot be bounded above by a natural number. It then follows that A^{-1} cannot be bounded above by any real number (why?).)

(b) For $c > 0$, let $cA = \{\frac{c}{2^n} : n \in \mathbb{N}\} = \{\frac{c}{2}, \frac{c}{4}, \frac{c}{8}, \frac{c}{16}, \frac{c}{32}, \dots\}$.
Combine a previous result with part (a) to show that $\inf cA = 0$. (*Hint*: By a previous result, if $c \geq 0$, then $\inf(cA) = c \inf A$.)